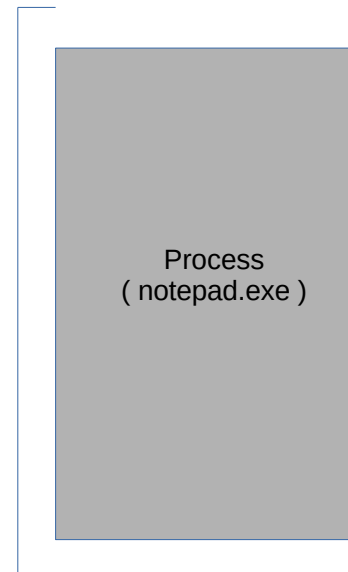


Creating Process

CreateProcess()



notepad.exe



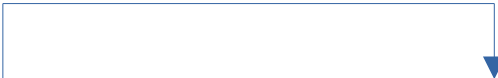
RAM

Function Prototype

```
BOOL CreateProcess(  
    lpApplicationName,    // Path to the executable  
    lpCommandLine,        // Command-line arguments  
    lpProcessAttributes,  // Security attributes for the process  
    lpThreadAttributes,   // Security attributes for the primary thread  
    bInheritHandles,      // Inherit handles from parent process  
    dwCreationFlags,      // Process creation flags  
    lpEnvironment,        // Pointer to environment block  
    lpCurrentDirectory,   // Working directory of the new process  
    lpStartupInfo,        // Pointer to STARTUPINFO structure  
    lpProcessInformation  // Pointer to PROCESS_INFORMATION structure  
);
```

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Flag	Description
CREATE_NEW_CONSOLE	Creates a new console window for the child process.
CREATE_SUSPENDED	<i>Creates the process in a suspended state. (Used in process injection.)</i>
CREATE_NO_WINDOW	Runs the process without showing any window (used for stealth malware).
CREATE_DEFAULT_ERROR_MODE	Uses the parent's error mode.
CREATE_BREAKAWAY_FROM_JOB	Allows the process to run outside of a job object.
CREATE_SEPARATE_WOW_VDM	Runs the process in a separate virtual DOS machine (for 16-bit apps).
CREATE_SHARED_WOW_VDM	Runs in a shared DOS environment (legacy stuff).
DEBUG_PROCESS	Allows the parent process to debug the child process.
DEBUG_ONLY_THIS_PROCESS	Only allows the parent to debug this process.
DETACHED_PROCESS	Runs the process without associating with the parent console.
EXTENDED_STARTUPINFO_PRESENT	Allows passing extended startup information.
INHERIT_PARENT_AFFINITY	Inherits the CPU affinity of the parent process.
CREATE_PROTECTED_PROCESS	Creates a protected process (used in anti-debugging).

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```



It controls how the new process starts
(window size, position, appearance, etc.).

```
typedef struct _STARTUPINFO {  
    DWORD cb;                // Size of the structure  
    LPSTR lpReserved;        // Reserved (always NULL)  
    LPSTR lpDesktop;         // Desktop name (NULL = default)  
    LPSTR lpTitle;           // Console title (NULL = default)  
    DWORD dwX, dwY;          // Window position (ignored if not set)  
    DWORD dwXSize, dwYSize;  // Window size  
    DWORD dwFlags;           // specify which flag is used (e.g., SW_HIDE)  
    WORD wShowWindow;        // Controls the window state (SW_SHOW, SW_HIDE, etc.)  
    ... (Other fields not commonly used)  
} STARTUPINFO;
```

Function Prototype

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    lpProcessInformation  // Pointer to PROCESS_INFORMATION structure  
);
```



Holds important information about the newly created process.

```
typedef struct _PROCESS_INFORMATION {  
    HANDLE hProcess;    // Handle to the process  
    HANDLE hThread;     // Handle to the main thread  
    DWORD dwProcessId;  // Process ID  
    DWORD dwThreadId;   // Thread ID  
} PROCESS_INFORMATION;
```